

ECON 211: Economic Statistics

Professor Vladyslav Nora

Spring 2025

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Class Hours: M/W/F, 11:00–11:50

Classroom: 8.327

Office Hours: W/F 13.30-14.30

TA: TBA

Course description

This course is an introduction to probability and statistics. We will learn how to think about risk and uncertainty in a *systematic and consistent way*. It is instrumental for your understanding of economic theory, econometrics, and generally any other modern scientific discipline that deals with uncertainty. The first part of the course (roughly the first two-thirds of the lectures) covers the theory of probability which is a mathematical tool to think about uncertainty. The second part covers basic statistics which applies probability theory to analyze real-world data.

Learning objectives

- Know and understand key concepts, techniques, and results from the textbooks and lectures.
- Be able to solve exercises in the textbooks and problem sets.
- Understand the applications and limitations of probability theory and statistics.
- Learn to use AI tools for simulations, problem solving, and studying in an ethical and transparent way.

Textbooks

There are two required textbooks:

1. (BH) *Introduction to Probability*, by Joseph K. Blitzstein and Jessica Hwang (2nd edition). A free online version of the second edition is available [here](#). The book is based on the author's lectures at Harvard, which are also available online.
2. (KB) *Statistics for Management and Economics*, by Gerald Keller (11th Edition).

Schedule

The detailed lecture schedule, problem set due dates, exam dates, additional materials and announcements are posted in real time on Moodle. Key dates for the Spring 2025:

- First class: January 12
- Spring break: March 2–9
- Midterm exam: March 11th
- Quiz 1: January 30th
- Quiz 2: March 30th
- Last class: April 24th
- Final exam: during the exam period (TBA)

How to study in this class

1. Practice is the single most important part of this course. Solving as many problems as you can on your own. Use AI tools (such as large language models like ChatGPT, Claude, Gemini etc.) to check your work after you have made a serious attempt. If you cannot solve a problem, ask AI first only for hints or partial guidance, not for the entire solution. This way you will actively test and deepen your understanding of the concepts discussed, instead of passively consuming answers. Remember: *believing that you understand something is not the same as actually understanding it*. Only by struggling with problems and finding solutions can you reliably measure your own comprehension.
2. Sources of practice problems:
 - Problem sets (6 total). These are mandatory and graded pass/fail. To pass, you must *attempt* most problems with hand-written solutions. Whether a problem is solved correctly does not affect grading; it is sufficient to demonstrate a genuine effort on each problem. Solutions to selected problems in problem sets will be distributed after the submission deadline. Each page must contain your name and ID number in the upper right corner. If you fail to submit more than two problem sets, you will not be admitted to the final exam.
 - Exercises in Keller's (any edition) and Blitzstein & Hwang's problem collections after the corresponding chapters.
 - You should try to use AI chatbots to give you practice questions, especially given that you can ask for exercises on a specific topic you feel like doing. For example: "Give me 2 simple, and 2 more advanced problems concerning the use of the law of total probability. The problems should be on the undergraduate course level for economics students."

Assessment

- Problem sets (6): Pass/Fail (must pass at least 4 of 6 to qualify for final exam).
- Quizzes (2): 35% (the first quiz weight is 15%, and the second quiz weight is 20%).
- Midterm exam: 25% (covers probability theory).
- Final exam: 40% (cumulative, covers entire course).

Exam rules: All exams are closed-book. No notes, calculators, or any electronic devices are permitted. Answers must be written in pen. Unless the exam rules specify otherwise, only final answers are graded, and they must be clearly circled.

Curving: Grades will be curved to approximately follow the distribution: 15% receive A/A-, 50% receive B+/B/B-, 20% receive C and 10% receive D+/D/F, with a median roughly around B- (2.67).

Attendance and Missed Exams

Attendance is strongly encouraged. Students with valid medical excuses must submit supporting documentation to the University Health Center (UHC) and notify the instructor. If a medical excuse is approved, the weight of the missed assessment will be redistributed proportionally across the remaining assessments. Without prior approval, a missed assessment receives a score of zero. There are no retakes, except of retakes of the final exam that will be arranged only in cases of certified medical emergencies. A student who retakes the final will be given an incomplete grade first.

AI policy

Except during exams, students are welcome to use generative AI tools in a totally unrestricted fashion, for any purpose, at no penalty. We believe that knowledge and effective use of these tools will be important for your future professional roles. However, you should note that all large language models tend to produce inaccurate and unreliable output. You will be responsible for any inaccurate, biased, offensive, or otherwise unethical content you submit regardless of whether it originally comes from you or an AI tool. If you use an AI tool, its contribution must be acknowledged by clearly acknowledging the extent of the AI tool usage by citing the source of the AI contribution or by completing a declaration of how AI has been used in the production of the assessment submission, included on the cover page or the beginning of the submission. Unacknowledged use of AI tools will be treated as plagiarism and penalties will be applied according to the NU Student Code of Conduct.

Misconduct and Academic Integrity

Academic misconduct (plagiarism, cheating, unacknowledged AI use) results in failure of the assignment or exam and may trigger further disciplinary action under the NU Code of Conduct. Violations of the exam rules lead to automatic failure of that exam.

Prerequisites

Students should review basic set theory, calculus (derivatives, integrals, geometric series), and proof techniques (contradiction, induction).

Students with special learning needs

The Nazarbayev University Special Learning Needs Committee (SLNC) is committed to creating an equitable and inclusive education environment for all students. If you have a qualified special learning need (physical, cognitive, socio-emotional, and psychological), please contact the SLNC as early as possible to ensure you receive the fullest support available.

If you already have approved SLNC accommodations, please share them with your teaching faculty as soon as possible. Accommodations cannot be applied retroactively, and will only be active once your teaching faculty has received them. This includes conditions that may be physical, cognitive, socio-emotional, and psychological.

For more information, please contact SLNC@nu.edu.kz and follow the link to the Accommodation Request Instructions.